ExpNo -15 Experiment to understand the data preprocessing in data science

Program:

```
import numpy as np
import pandas as pd
df = pd.read_csv("pre-process_datasample.csv")
df
df.info()
df.Country.mode()
df.Country.mode()[0]
type(df.Country.mode())
# Fill missing values
df['Country'] = df['Country'].fillna(df['Country'].mode()[0])
df['Age'] = df['Age'].fillna(df['Age'].median())
df['Salary'] = df['Salary'].fillna(round(df['Salary'].mean()))
df
# One-hot encoding
pd.get_dummies(df.Country)
updated_dataset = pd.concat(
   [pd.get_dummies(df.Country), df.iloc[:, [1, 2, 3]]],
   axis=1
)
updated_dataset
```

Output:

<class 'pandas.DataFrame'>

RangeIndex: 10 entries, 0 to 9

Data columns (total 4 columns):

#	Column	Non-Null Count	Dtype
0	Country	8 non-null	str
1	Age	8 non-null	float64
2	Salary	8 non-null	float64
3	Purchased	10 non-null	str

dtypes: float64(2), str(2)

memory usage: 452.0 bytes

	France	Germany	Spain	Age	Salary	Purchased
0	True	False	False	44.0	72000.0	No
1	False	False	True	27.0	48000.0	Yes
2	False	False	True	30.0	54000.0	No
3	False	False	True	38.0	61000.0	No
4	False	True	False	37.5	58000.0	Yes
5	True	False	False	35.0	61375.0	Yes
6	False	False	True	37.5	52000.0	No
7	False	False	True	48.0	79000.0	Yes
8	False	True	False	50.0	61375.0	No
9	False	False	True	37.0	67000.0	Yes